

GreenPredict-AI

Environmental Plant Survival Prediction & Recommendation System

Project Documentation

1. Introduction

What is GreenPredict-AI?

GreenPredict-AI is an artificial intelligence powered plant survival prediction and recommendation system designed specifically for Maharashtra, India. It uses machine learning to analyze local soil conditions and city-specific climate data to predict the likelihood of successful growth for eight native tree species.

The system is built as a web application that provides instant, data-driven guidance to anyone involved in tree plantation. By entering a few soil measurements and selecting a city, users receive a ranked list of tree species along with their predicted survival probabilities — all without requiring any botanical expertise.

GreenPredict-AI transforms complex agricultural science into simple, actionable recommendations that can be used directly in the field.

Who Is It For?

GreenPredict-AI is designed for:

- Government forest departments running afforestation programs
- NGOs and environmental organizations planning plantation drives
- Community groups and resident welfare associations
- Schools and colleges creating green campuses
- Farmers integrating trees into agricultural land
- Highway and roadside plantation coordinators
- Environmental volunteers working in Maharashtra

2. Problem Statement

The Challenge of Tree Plantation in Maharashtra

Maharashtra is one of India's largest states with highly diverse geography, soil types, and rainfall patterns. Annual rainfall ranges from as low as 547 mm in Solapur to over 2100 mm in Mumbai. Despite this diversity, tree plantation programs across the state frequently fail because species are selected without considering local conditions.

Core Problems

- Plantation drives suffer from failure rates estimated between 40 and 60 percent due to poor species selection
- Planters choose trees based on availability, appearance, or tradition rather than soil and climate compatibility
- No accessible tool exists for field workers to evaluate and compare multiple species simultaneously Government plantation guidelines are generic and not tailored to specific city or district conditions

- Farmers and NGO volunteers have no access to expert botanical advice in the field
- Wasted resources including money, water, labor, and time are invested in saplings that will not survive
- Failed plantation programs discourage future greening initiatives and reduce public confidence

The Consequence

When the wrong tree is planted in the wrong soil, the sapling dies within months. The funds spent on purchase, transport, planting, and watering are entirely wasted. On a large scale, this means thousands of trees planted during government drives fail silently, and the intended environmental benefits never materialize.

The Solution

GreenPredict-AI solves this problem by providing an instant, science-based recommendation for which tree species will thrive under a given set of soil and climate conditions. It puts expert botanical knowledge in the hands of every field worker through a simple web interface.

3. System Overview

What the System Does

GreenPredict-AI accepts nine input parameters describing the soil at a planting site and the climate of the selected city. It then uses a trained machine learning model to predict the survival probability for all eight supported tree species simultaneously, ranks them from highest to lowest, and displays the results clearly on screen.

Eight Supported Tree Species

The system supports the following native Maharashtra tree species:

- Neem (*Azadirachta indica*) — drought tolerant, medicinal, pest control, air purification
- Banyan (*Ficus benghalensis*) — India's national tree, wide canopy, wildlife habitat
- Peepal (*Ficus religiosa*) — releases oxygen around the clock, sacred, roadside tree
- Mango (*Mangifera indica*) — India's national fruit tree, dense shade, orchards
- Jamun (*Syzygium cumini*) — medicinal fruit, beneficial for diabetes, iron-rich berries
- Tamarind (*Tamarindus indica*) — drought resistant, edible pods, suitable for dry regions
- Arjun (*Terminalia arjuna*) — Ayurvedic heart medicine, thrives near riverbanks
- Gulmohar (*Delonix regia*) — brilliant red-orange flowers, pollution tolerant, urban tree

System Components

The system consists of three main parts working together:

- **Web Application** — A modern, responsive interface built with React.js where users enter inputs and view results
- **Prediction Engine** — A Python backend running a trained Random Forest machine learning model
- **AI Chatbot** — A Retrieval-Augmented Generation chatbot that answers questions about trees, soil, and the system using the project documentation as its knowledge source

4. How the System Works

Complete Workflow

Phase One — Data Preparation

Since labeled real-world data for Maharashtra plant survival was not publicly available, training data was created using verified botanical parameters for each tree species. Each species has known preferences for rainfall, temperature, soil pH, and soil type.

- Rainfall and temperature values were generated using a statistical normal distribution centered at each species ideal range
- This ensures most generated samples represent near-optimal conditions, making the model learn realistic patterns
- Soil type was randomly selected from each species list of compatible soils
- Nutrient values like nitrogen, phosphorus, and potassium were generated across realistic agricultural ranges
- Four hundred samples were created for each of the eight species, producing a total dataset of 3200 rows

Phase Two — Model Training

- The dataset is loaded and the soil type column is converted from text into numeric format through one-hot encoding
- Input features are separated from the target label which is the plant species name
- The data is split into 80 percent for training and 20 percent for testing
- A Random Forest Classifier with 300 decision trees is trained on the training data
- The trained model is evaluated using a confusion matrix and classification report
- The model is saved to disk so it can be loaded instantly without retraining

Phase Three — User Prediction

- The user opens the web application and navigates to the Predict page
- The user selects their Maharashtra city, which automatically retrieves the annual rainfall and average temperature for that city
- The user enters seven soil parameters from their soil test report
- The user selects the tree species they are considering planting
- The system assembles all nine parameters into the format the model expects
- The model runs the input through all 300 decision trees and collects their votes
- The final survival probability for each species equals the percentage of trees that voted for it
- Results are displayed as a ranked list of all eight species with survival percentages

Phase Four — Recommendation

- The system compares the user's chosen tree against all predictions
- If the chosen tree has the highest survival probability, a success confirmation is shown
- If another tree has a higher probability, a recommendation alert highlights the better option
- The user can download a PDF report of the full results for documentation or sharing

5. Input Parameters

Overview

The system requires nine input parameters to generate a prediction. Seven are entered manually by the user. Two are automatically retrieved from the built-in climate database when the user selects their city.

Soil Parameters — Entered by User

Soil Type

The physical composition of the soil at the planting site. Soil type determines how well water drains, how much air reaches roots, and how many nutrients the soil retains.

Available options:

- Sandy — drains quickly, low nutrient retention, suitable for drought-tolerant species
- Loamy — ideal balance of drainage and nutrients, suitable for most tree species
- Alluvial — rich and fertile, found near rivers, excellent for tall shade trees
- Lateritic — reddish, found in high-rainfall areas, suits species that tolerate leaching
- Red Loam — well-drained, suitable for drought-resistant species like Tamarind

Soil pH

A measure of soil acidity or alkalinity on a scale from 0 to 14. A pH of 7 is neutral. Values below 7 are acidic and values above 7 are alkaline.

- Most trees prefer a pH between 6.0 and 8.0
- Acidic soils can lock up nutrients and harm roots
- Highly alkaline soils reduce iron and manganese availability
- Measured using a pH meter or a soil test kit available at agricultural offices

Nitrogen

The concentration of nitrogen in the soil, measured in kilograms per hectare. Nitrogen is the primary nutrient responsible for leaf and stem growth. Deficiency causes yellowing of leaves and stunted growth. Typical values in Indian agricultural soils range from 100 to 500 kg per hectare.

Phosphorus

The concentration of phosphorus in the soil, measured in kilograms per hectare. Phosphorus supports root development, flowering, and the plant's ability to store and transfer energy. Typical values range from 5 to 60 kg per hectare.

Potassium

The concentration of potassium in the soil, measured in kilograms per hectare. Potassium improves drought tolerance, disease resistance, and the quality of fruit produced. Typical values range from 50 to 400 kg per hectare.

Organic Carbon

The percentage of organic matter present in the soil. Organic carbon is an indicator of soil health, biological activity, and long-term fertility.

- Values above 0.75 percent indicate healthy soil
- Values below 0.5 percent indicate degraded or depleted soil
- Measured as a percentage, typically between 0.3 and 1.2 in Indian soils

Electrical Conductivity

A measure of soil salinity, reported in decisiemens per meter. High salinity interferes with a plant's ability to absorb water and nutrients.

- Values below 2.0 dS/m are safe for most tree species
- Values above 4.0 dS/m are considered highly saline and harmful
- Measured using an EC meter as part of a standard soil test

Climate Parameters — Auto-Fetched from City Database

Annual Rainfall

The total amount of rainfall the selected city receives per year, measured in millimeters. This is the single most important factor in determining which tree species will survive. The system automatically retrieves this value when the user selects their city.

Average Temperature

The average annual temperature of the selected city, measured in degrees Celsius. Temperature determines which species can survive local thermal conditions throughout the year. This is also automatically retrieved by city selection.

Maharashtra City Climate Reference

City	Annual Rainfall	Average Temperature
Pune	722 mm	26°C
Mumbai	2167 mm	27°C
Nagpur	1205 mm	28°C
Nashik	690 mm	25°C
Aurangabad	726 mm	27°C
Solapur	547 mm	29°C
Kolhapur	1100 mm	25°C
Satara	935 mm	25°C
Amravati	920 mm	28°C
Nanded	883 mm	28°C

6. Machine Learning Model

What is a Random Forest?

A Random Forest is a machine learning algorithm that works by building a large number of decision trees and combining their results. Each decision tree in the forest is like an independent expert who examines the input data and makes a prediction. The final output is determined by counting how many trees voted for each option — the option with the most votes wins.

The strength of Random Forests comes from this combination. While a single decision tree might make mistakes, the collective opinion of 300 trees is far more reliable and accurate.

Why Random Forest for This Project?

Random Forest was chosen for GreenPredict-AI for several reasons:

- It handles both numeric inputs like pH and temperature and categorical inputs like soil type naturally
- It is robust to noise in training data and does not require data to follow any particular distribution
- It outputs a probability for each class rather than just a single answer, which is exactly what this system needs
- It provides feature importance scores that reveal which inputs matter most for prediction
- It performs excellently on small to medium datasets like the 3200 samples used here

- It is interpretable enough for a scientific project while being highly accurate

Model Configuration

The model was configured with the following settings:

- Number of trees: 300 decision trees (a higher number improves stability and accuracy)
- Maximum tree depth: 15 levels (limits how specific each tree can be, preventing overfitting)
- Minimum samples to split: 5 (a node must have at least 5 samples before it can split further)
- Output type: Probability scores for all eight species simultaneously
- Training accuracy: Typically between 90 and 97 percent on the held-out test set

How the Model Makes a Prediction

When a user submits their inputs, the following happens inside the model:

1. The nine input values are formatted exactly as the model expects
2. The input travels through all 300 decision trees one by one
3. Each tree examines the inputs and arrives at its own prediction for which species would best survive
4. The votes from all 300 trees are counted for each of the eight species
5. The survival probability for each species equals the fraction of trees that voted for it
6. For example, if 270 out of 300 trees vote for Neem, the Neem survival probability is 90 percent

Feature Importance

After training, the model reveals which input parameters contributed most to its predictions. Listed from most to least important:

1. Rainfall — the dominant factor, determines if water needs of each species are met
2. Temperature — separates tropical species from those with wider thermal tolerance
3. Soil pH — determines nutrient availability and chemical compatibility
4. Soil Type — affects drainage, root penetration, and water retention
5. Organic Carbon — indicator of overall soil biological health
6. Nitrogen — primary growth nutrient, affects leafy biomass
7. Phosphorus — root development and reproductive health
8. Potassium — stress tolerance, disease resistance
9. Electrical Conductivity — salinity stress indicator

Species Survival Requirements

Each species in the model has a defined range of conditions under which it thrives:

- Neem: rainfall 450 to 1150 mm, temperature 20 to 40°C, pH 6.0 to 8.5, Sandy or Loamy soil
- Banyan: rainfall 500 to 4000 mm, temperature 9 to 40°C, pH 6.0 to 8.5, Alluvial or Loamy soil
- Peepal: rainfall 600 to 3000 mm, temperature 10 to 40°C, pH 6.0 to 8.5, Alluvial or Loamy soil
- Mango: rainfall 750 to 1900 mm, temperature 15 to 40°C, pH 5.5 to 7.5, Loamy soil only
- Jamun: rainfall 700 to 3000 mm, temperature 10 to 40°C, pH 6.0 to 9.0, Alluvial or Lateritic soil
- Tamarind: rainfall 750 to 1900 mm, temperature 15 to 40°C, pH 5.5 to 6.8, Red Loam or Sandy soil
- Arjun: rainfall 750 to 1900 mm, temperature 20 to 35°C, pH 6.5 to 7.5, Alluvial soil only

- Gulmohar: rainfall 700 to 1200 mm, temperature 14 to 35°C, pH 6.0 to 8.0, Sandy or Loamy soil

7. Prediction & Recommendation System

How Results Are Displayed

After the user submits their inputs, the system displays:

- The selected tree's survival probability as a large percentage
- A rating label indicating compatibility level
- A ranked list of all eight tree species ordered from highest to lowest survival probability
- A recommendation alert if a more suitable species was identified

Survival Rate Interpretation

Survival Rate	Meaning
85% and above	Excellent compatibility. Tree will very likely thrive with no special care needed.
70% to 84%	Good potential. Tree should grow well with standard maintenance and watering.
55% to 69%	Moderate compatibility. Tree may survive but needs extra care and monitoring.
Below 55%	Low compatibility. This species is not well suited for these conditions.

Recommendation Logic

The recommendation system works as follows:

- After prediction, the system identifies which of the eight species received the highest survival probability
- If the user's selected tree is also the top-scoring species, a green confirmation message is shown confirming it is the optimal choice
- If a different species scored higher, a yellow alert appears showing the better recommendation with its survival probability
- The user can then choose to plant the recommended species or proceed with their original choice
- All eight species are displayed as ranked cards so the user can compare all options side by side

PDF Report

The system generates a downloadable PDF report containing:

- All soil parameters entered by the user
- The selected city and its climate data
- The selected tree's survival probability and rating
- All eight species ranked by survival probability
- The best recommendation if different from the user's selection

This report can be used for documentation, funding applications, or sharing with program coordinators.

8. Key Features

Prediction Features

- Simultaneous survival probability prediction for all eight native tree species
- Automatic retrieval of rainfall and temperature data from the city climate database
- Real-time ranking of all species from most to least compatible
- Automatic detection of a better alternative and display of a recommendation alert
- Downloadable PDF report of the complete prediction results

Information Features

- Tree Encyclopedia with detailed profiles of all eight supported species
- Each profile includes soil requirements, climate preferences, key benefits, and description
- Search functionality to find trees by name or characteristics
- Detailed species popup with full information

Chatbot Features

- AI-powered chatbot that answers questions about trees, soil conditions, and the system
- Chatbot uses only the project documentation as its knowledge source
- Powered by Retrieval-Augmented Generation ensuring answers are accurate and grounded
- No hallucination — the chatbot will not invent information not present in the source document
- Answers questions about specific tree species, Maharashtra cities, soil parameters, and how to use the system

User Experience Features

- Clean, modern, green-themed interface reflecting the environmental purpose
- Fully responsive design that works on desktop and mobile devices
- Form validation ensuring all required fields are filled before prediction runs
- Loading indicators and smooth animations throughout the interface
- Clear visual feedback distinguishing optimal and suboptimal recommendations

9. Use Cases

Government Plantation Programs

Maharashtra's forest department and municipal corporations run annual plantation drives planting thousands of trees across districts. GreenPredict-AI can be used by program officers to determine the most suitable species for each location before purchasing saplings. By entering soil test data from target locations, officers can ensure every tree planted has the highest possible chance of survival, maximizing the return on government investment.

Benefit: Significantly reduces the 40 to 60 percent failure rate in government drives.

NGO Environmental Projects

Environmental NGOs often plan large-scale plantation campaigns in rural and tribal areas where botanical expertise is not locally available. GreenPredict-AI gives NGO field coordinators the ability to make science-based species decisions using only a soil test kit and the web application.

Benefit: Allows organizations without in-house ecologists to make expert-level planting decisions.

Community Greening Initiatives

Resident welfare associations, housing societies, and local community groups frequently want to plant trees in their neighborhoods but lack guidance on suitable species. GreenPredict-AI provides instant recommendations that any community member can obtain without technical knowledge.

Benefit: Empowers ordinary citizens to participate in greening initiatives with confidence.

School and College Campus Greenery

Educational institutions across Maharashtra are creating green campuses as part of environmental awareness programs. GreenPredict-AI allows science teachers and campus administrators to select trees that will genuinely thrive in their specific location, making the exercise educational and successful.

Benefit: Combines environmental education with practical, data-driven decision making.

Agricultural Agroforestry

Farmers in Maharashtra are increasingly adopting agroforestry models that integrate trees with crops. Choosing the right tree species is critical because it must be compatible with existing soil chemistry without competing harmfully with crops. GreenPredict-AI helps farmers identify tree species that complement their soil conditions.

Benefit: Supports sustainable farming by enabling scientifically selected agroforestry integration.

Highway and Roadside Plantation

The Public Works Department and highway authorities plant trees along roads for shade, erosion control, and pollution reduction. Roadside conditions often involve disturbed soil, compacted ground, and pollution stress. GreenPredict-AI identifies pollution-tolerant and resilient species like Gulmohar and Neem that are specifically suited for such conditions.

Benefit: Ensures roadside trees survive urban stress conditions and provide lasting benefit.

Dry Region Plantation Planning

Districts in Maharashtra's drought-prone Marathwada and Vidarbha regions receive minimal annual rainfall. Standard plantation approaches fail because the trees selected are not drought tolerant. GreenPredict-AI automatically weights rainfall compatibility heavily and consistently recommends appropriate drought-tolerant species like Neem, Tamarind, and Gulmohar for these regions.

Benefit: Addresses the specific challenges of planting in Maharashtra's water-scarce districts.

Environmental Impact Assessment

Before any development project that affects green cover, environmental impact assessments require a plan for compensatory plantation. GreenPredict-AI can generate site-specific species recommendations supported by soil data, making these assessments more credible and ensuring the compensatory trees actually survive.

Benefit: Adds scientific credibility to environmental compliance documentation.

GreenPredict-AI — Bringing Artificial Intelligence to Environmental Plantation Planning in Maharashtra

Making every tree count.